

Modeling cash flows using Linear Programming

1 Introduction

In these lecture notes, we will see two problems arising in finance that can be modeled using Linear Programming. Later in the course, we will see that other problems arising in finance such as portfolio optimization problems can be modeled using non-linear (convex) optimization.

2 A cash flow problem

We have \$100 to invest. There are three investment vehicles.

- a. for every \$1 invested now (that is, at the beginning of year 0), we get 0.1 one year from now (that is, at the beginning of year 1), and \$1.3 three years from now.
- b. for every \$1 invested now, we get 0.2 one year from now, and \$1.1 two years from now.
- c. for every \$1 invested a year from now, we get 1.5 three years from now.

Alternatively, in every year we can deposit money in a money-market account (mma), with an interest rate of 2% per year.

Goal: find an investment strategy so that we never overdraw and we maximize money available at the beginning of year 3.

First: we model the decisions to be made using **variables**:

- x_a, x_b, x_c : dollars invested in each option.
- y_0, y_1, y_2, y_3 : dollars available in year 0, 1, 2, 3, after all returns on investments / payments have been processed. Note that in years 0, 1, 2 those money will be invested in the mma.

Second: Write the **objective function**

$$\max y_3$$

Third: Write the **constraints** connecting the variables

$$y_0 = 100 - x_a - x_b \quad (\$ \text{ available in year 0} = \text{original budget minus money invested in } a, b)$$

$$y_1 = 0.1x_a + 0.2x_b + 1.02y_0 - x_c \quad (\$ \text{ available in year 1} = \text{returns from investments } a, b \text{ plus } 2\% \text{ increase on money in mma in year 0, minus money in investment } c)$$

$$y_2 = 1.1x_b + 1.02y_1 \quad (\$ \text{ available in year 2} = \text{returns from investment } b \text{ and } 2\% \text{ increase on money in mma in year 1})$$

$$y_3 = 1.3x_a + 1.5x_c + 1.02y_2 \quad (\$ \text{ available in year 3} = \text{returns from inv. } a, c \text{ and } 2\% \text{ increase on money in mma in year 2})$$

What else?

$$x_a, x_b, x_c, x_d, y_0, y_1, y_2, y_3 \geq 0$$

A general formulation

Let us consider the more general version of the problem seen above. We are given:

- Budget b ;
- Time horizon T ;
- n investment vehicles;
- A table Q gives, for each year t ($t = 0, 1, \dots, T$) and investment vehicle j the normalized amount of money j returns (or requires) in year t .
- Money in a money-market account gives a yearly return of $r\%$.

The goal is to decide investments so that we never overdraw and maximize the money available at the beginning of year T .

In the previous example, we had $b = 100, T = 3, r = 0.02$, and

$$Q = \left(\begin{array}{c|ccc} & a & b & c \\ \hline 0 & -1 & -1 & 0 \\ 1 & 0.1 & 0.2 & -1 \\ 2 & 0 & 1.1 & 0 \\ 3 & 1.3 & 0 & 1.5 \end{array} \right)$$

where rows denotes the years and columns denotes the investments. For instance, the entry -1 in row 0, column a says that investment a requires us to pay 1 dollar in year 0. Similarly, row 1, column a says that investment a will return 0.1 dollars in year 1. These values are normalized in the sense that if we invest k dollars in investment a in year 0, then our return will be $k \cdot 0.1$ dollars in year 1 and $k \cdot 1.3$ dollars in year 3.

Let us now formulate this general problem using linear programming.

First we model the variables. We let $x_1, x_2, \dots, x_n$ be the money invested in investment $1, 2, \dots, n$. We arrange them in a vector

$$x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}.$$

Then we let $y_0, y_1, y_2, \dots, y_T$ be the money available at the beginning of year $0, 1, 2, \dots, T$ after all returns on investments / payments have been processed. Again, we arrange them in a vector

$$y = \begin{pmatrix} y_0 \\ y_1 \\ y_2 \\ \vdots \\ y_T \end{pmatrix}.$$

Recall that in years $0, 1, \dots, T - 1$, those money will be deposited in the mma.

Second we write the objective function

$$\max y_T$$

Third we write the constraints. We start with nonnegativity:

$$x, y \geq 0$$

(since we can only invest a positive amount of money and we do not want to overdraft). Let's write the constraint for year 0:

$$y_0 = b + Q_{0,1}x_1 + Q_{0,2}x_2 + \dots, Q_{0,n}x_n, \quad (1)$$

where for $t = 1, \dots, T$, we have that $Q_{0,t}$ is the entry corresponding to row 0, column t in the matrix Q . (1) can be written compactly as

$$y_0 = b + \sum_{j=1}^n Q_{0,j}x_j, \quad (2)$$

In the following, let $Q_{i,t}$ be the entry in row i , column t of the matrix Q . For the generic year t , the constraint looks as follows

$$y_t = Q_{t,1}x_1 + Q_{t,2}x_2 + \dots, Q_{t,n}x_n + (1 + r)y_{t-1},$$

where the last term is due to the return on the money invested in the mma. Again, it can be written compactly as

$$y_t = \sum_{j=1}^n Q_{t,j}x_j + (1 + r)y_{t-1}. \quad (3)$$

Thus, we can write the LP as

$$\begin{array}{ll} \max & y_T \\ \text{s.t} & (2), (3) \text{ hold} \\ & x, y \geq 0 \end{array}$$

Observe moreover that the right-hand side of the constraint from (2), (3) corresponding to year t is given by the product of the t -th row of matrix Q with the x vector plus b (when $t = 0$) or $(1 + r)$ times the $(t - 1)$ -th entry of vector y . Thus, if we denote by $\bar{y}$ the vector defined as

$$\bar{y} = \begin{pmatrix} b/(1 + r) \\ y_0 \\ y_1 \\ y_2 \\ \vdots \\ y_{T-1} \end{pmatrix},$$

we can write the LP in matrix form as:

$$\begin{array}{ll}
\max & y_T \\
\text{s.t} & Qx + (1+r) \cdot \bar{y} = y \\
& x, y \geq 0
\end{array}$$

3 A short-term financing problem

The expected cash flows (in thousand of dollars) of the company in the next months are the following

Months	1	2	3	4	5	6
Net cash flow	-150	-100	200	-200	50	300

The company has the following sources of funds:

- Credit of up to 100,000 at an interest rate of 1% per month;
- Additional debt for a 3-month period at an interest rate of 2% per the 3-month periods (this money can be accessed in each of the months 1,2,3);

Each month, excess funds can be invested at an interest rate of 0.3% per month. We want to formulate the problem of maximizing the excess funds of the company after 6 months.

We define the following variables:

- $x_1, x_2, \dots, x_5$ = money borrowed through the first credit line in months 1, 2, ..., 5 (in thousands of dollars)
- y_1, y_2, y_3 = money borrowed through the second credit line in months 1, 2, ..., 3 (in thousands of dollars)
- $z_1, z_2, \dots, z_6$ = excess money in month 1, 2, ..., 6 (in thousands of dollars)

Using those, the Objective function becomes

$$\max z_6$$

The constraints relating the various variables are as follows

$$\begin{aligned}
x_1 + y_1 - z_1 &= 150, \\
x_2 + y_2 - 1.01 x_1 + 1.003 z_1 - z_2 &= 100, \\
x_3 + y_3 - 1.01 x_2 + 1.003 z_2 - z_3 &= -200, \\
x_4 - 1.02 y_1 - 1.01 x_3 + 1.003 z_3 - z_4 &= 200, \\
x_5 - 1.02 y_2 - 1.01 x_4 + 1.003 z_4 - z_5 &= -50, \\
-1.02 y_3 - 1.01 x_5 + 1.003 z_5 - z_6 &= -300, \\
x_j &\leq 100 \quad \text{for } j = 1, \dots, 5, \\
x_j &\geq 0 \quad \text{for } j = 1, \dots, 5, \\
y_j &\geq 0 \quad \text{for } j = 1, \dots, 3, \\
z_j &\geq 0 \quad \text{for } j = 1, \dots, 5.
\end{aligned}$$

Note in each of the constraints (other than the nonnegative variables), the right-hand side is given by the opposite of the cash flow for the corresponding month (the first one is the opposite of the cash flow in the first month, the second is the opposite of the cash flow in the second month, etc.). The first constraint tell us that:

- the money we borrow through the first credit line (x_1), plus
- the money we borrow through the second credit line (y_1), minus
- the excess money (z_1 , that we will invest)

must equal the money requested by the cash flow table in the first month.

The second constraint is similar, though we will have to pay back money borrowed through the first credit line in month 1 ($-1.01x_1$) and we will get 0.3% return on money invested in month 1 ($+1.003z_1$).

As for the other constraints, they are similar, but note, e.g., that in the constraint corresponding to month 4 we will have to pay back money borrowed through the second credit line in month 1 ($-1.02y_1$) and cannot borrow through the second credit line anymore (since $4 + 3 = 7$, and the timeline stops in month 6). Similarly, no money at all can be borrowed in month 6.